Projection Obstruction for Factorized States: Nonlinear Dynamics, Conditional ℓ^1 Diagnostics, and Finite-Time Defect Bounds

Jeremy H. Carroll

May 2026
Version: v5.0 Pre-Print (Conditional Diagnostics and Dependency Clarification)

Abstract

Projection-based approximations replace correlated states with product states through a nonlinear retraction. This paper studies the resulting discrepancy as an observer-relative obstruction: a finite coboundary defect measured by a declared replica-extensive, edge-decomposable diagnostic. The results are finite-dimensional, finite-time, and conditional on explicit observer assumptions.

We analyze this discrepancy in a Banach presheaf framework and establish three results. **(1)** For a C^2 retraction Π onto a submanifold M, the induced projected generator is generally nonlinear in the ambient affine coordinates; under a non-degeneracy condition on the Jacobian this rules out affine-linear representation in those coordinates (Theorem 3.1). This is not a claim excluding all possible nonlinear smooth conjugacies. **(2)** By formalizing Replica Extensivity plus edge decomposability across independent topological partitions, we derive independent measurement additivity as an observer assumption rather than a hidden modeling choice. Under these bounds, the ℓ^1 edge-sum diagnostic is selected up to edge weights, with uniformity requiring edge-transitivity or normalization (Theorem 4.2). **(3)** For finite-range, finite-dimensional lattice systems with non-commuting interactions, the induced defect exhibits non-vanishing density over a finite time interval independent of system size, as constrained by Lieb–Robinson bounds (Theorem 5.1).

Axioms RE1–RE5 are treated as structural observer bounds. Within the class of replica-extensive, edge-decomposable, symmetry-compatible observers, an ℓ^1 edge-sum coboundary diagnostic is selected; global unweighted form requires explicit edge normalization.

Crucially, **the mapping of this geometric defect into continuous $L^2_{\mathbf{RMS}}$ statistics (the Hilbert extension) is formally classified as a Conditional Minimality boundary**. The L^2 envelope is selected only after additional assumptions such as complex phase, bilinearity, reversibility, phase symmetry, and positive mass recovery are injected.

We further observe that RE1–RE5 are not consequences of graph structure alone: they characterize a separable observer regime. This grounds the ℓ^1 diagnostic conditionally, not universally.

These results formalize the geometric, categorical, and dynamical structure of projection-induced errors in factorized many-body approximations. The framework is finite-dimensional and finite-time; no claims are made regarding asymptotic or thermodynamic limits.

---

1 Introduction

Projection-based approximations are central to many-body physics and computation. Methods such as mean-field theory, Hartree–Fock, and product-state variational ansatze replace a correlated state with a product of local states, thereby reducing an exponentially large description to a tractable one. This replacement is implemented by a nonlinear map — formally, a retraction

— onto a low-dimensional manifold of factorized states. While the practical success of these methods is well established, the structural effect of this retraction on the underlying dynamics is less clearly understood: what information is systematically discarded, how it propagates, and how it should be measured relative to a declared observer class.

In this work, we isolate and analyze the discrepancy introduced by projection as a geometric and combinatorial object. We show that composing a linear many-body evolution with a nonlinear retraction produces a projected generator that is generally nonlinear in ambient affine coordinates, and that the resulting local inconsistencies assemble into a presheaf coboundary. Under explicit locality and observer symmetry conditions, this leads to a distinguished ℓ^1 diagnostic for projection-induced defects, and for finite-range systems we prove that this defect persists with non-vanishing density over a finite time interval independent of system size.

The goal is not to propose a new physical theory, but to provide a precise structural account of the error introduced by factorization.

We prove three structural theorems:

1. **Geometric Obstruction (Theorem 3.1):** The projected generator exhibits affine-coordinate nonlinearity; the theorem does not exclude arbitrary nonlinear smooth conjugacies.

2. **Conditional Additive Geometry (Theorem 4.2):** Under Replica Extensivity plus edge decomposability and normalization (RE1-RE5), the ℓ^1 coboundary norm is selected. Without edge normalization, the result is weighted ℓ^1. Relaxing the observer assumptions admits other norms.

3. **Density Persistence (Theorem 5.1):** The per-bond obstruction density is $\Omega(1)$ in system size over a finite local time window, with constants depending only on local dimension, interaction strength, active bond fraction, edge-density bound, the uniform local non-degeneracy bound, and the declared bounded-degree/Lieb–Robinson constants.

Scope. The framework applies rigorously to Banach presheaves over finite posets. We avoid speculative physics or infinite-time limits; the density scaling bound is finite-time only.

Conventions. Norms are denoted $\|\cdot\|$ with appropriate subscripts; scalar absolute values are denoted $|\cdot|$.

1.1 Relation to Companion Work

The additive diagnostic used throughout this paper — the ℓ^1 coboundary norm — is not introduced ad hoc. It is the output of a two-stage classification program establishing conditional ℓ^1 rigidity at progressively richer structural levels:

- **Stage 1 (combinatorial):** In "Coordinate-Wise Additivity and the ℓ^1 Norm on Finite Graph Cochains" [8], local additivity over disjoint coordinate supports and edge separability on finite graph cochains select an ℓ^1 geometry within the stated observer class. No Banach space structure or dynamics are required.

- **Stage 2 (functional-analytic):** In "The Free ℓ^1 Seminorm on Banach Presheaf Coboundaries" [6], the same rigidity persists when the cochain spaces are replaced by Banach presheaves and coordinate additivity is replaced by structural compatibility (functoriality under bounded linear maps).

By the classification results of these preceding papers, any diagnostic satisfying locality, additivity over disjoint supports, and structural compatibility with the underlying linear maps must be a weighted ℓ^1 coboundary norm, becoming proportional to the unweighted norm under

edge-uniformity assumptions. Accordingly, the defect quantity analyzed here is determined only relative to those observer constraints.

Although numbered Paper 001 in the preprint series, this paper uses the two companion rigidity results above as technical inputs. The numbering reflects publication order, not logical dependency.

The present paper is the dynamical stage of that classification: it shows that projection-induced dynamics naturally generate nontrivial coboundary defects, and that these defects persist with non-vanishing density under finite-range evolution. Together, the three papers characterize the replica-extensive, additive observer regime. Alternative observer classes correspond to different admissible transformation monoids and yield distinct obstruction geometries.

Conceptual Overview

```
Full correlated state space X
        |
        |  C^2 retraction Pi
        v
Product-state manifold M
        |
        |  induced curvature term D^2Pi(Am, .)
        v
Projected dynamics S_t = Pi . T_t|_M
        |
        |  local incompatibilities across regions
        v
Coboundary defects delta_e(s)
        |
        |  additive aggregation (Axioms 1-4)
        v
Total obstruction  I(s) = Sum ||delta_e(s)||
```

2 Structural Framework

We work with Banach-valued presheaves over finite posets, following standard categorical conventions [2].

2.1 Presheaves over Finite Posets

Let $(P, \leq)$ be a finite poset with covering relations $E_{\text{cov}}(P) = \{(a, b) : a < b,\ \nexists\, c \text{ with } a < c < b\}$. A **Banach presheaf** F over P assigns to each $x \in P$ a Banach space $F(x)$ and to each $(a, b) \in E_{\text{cov}}$ a contractive linear restriction map $r_{b,a} : F(b) \to F(a)$.

2.2 The Coboundary

For each edge $e = (a, b)$, the **coboundary** is $\delta_e(s) = r_{b,a}(s(b)) - s(a) \in F(a)$. A section descends if $\delta_e(s) = 0$ for all e. The ℓ^1 coboundary norm is $I(s) = \sum_{e \in E_{\text{cov}}} \|\delta_e(s)\|$.

2.3 Linear Semigroups and Projected Dynamics

A C_0 **contraction semigroup** on a Banach space X is $\{T_t\}_{t \geq 0}$ with generator $(A, \mathcal{D}(A))$. A C^2 **retraction** is a C^2 idempotent $\Pi : X \to M \subset X$ with M a regular C^2 submanifold. If $\{T_t\}$ does not preserve M, we define the **projected semigroup** $S_t = \Pi \circ T_t|_M$ mapping $M \to M$ locally.

3 Theorem I: Projection-Induced Affine Nonlinearity

3.1 Statement

Theorem 3.1 (Projection-Induced Affine Nonlinearity)**.** *Let X be a Banach space, $\{T_t\}$ a C_0 contraction semigroup with generator $(A, \mathcal{D}(A))$, and $\Pi : X \to M \subset X$ a C^2 retraction onto a C^2 submanifold $M \subset \mathcal{D}(A)$. Suppose:*

(a) (Non-invariance) There exists $t > 0$ such that $T_t(M) \not\subset M$.

(b) (Non-degeneracy) The induced vector field $B(m) = D\Pi(m)Am$ possesses a non-constant Jacobian.

*Then the projected generator $B(m) = D\Pi(m)Am$ is not affine-linear in any induced affine coordinate chart satisfying hypothesis (b). Consequently, the projected dynamics cannot be represented in those coordinates by an exact linear semigroup generator. This establishes coordinate-level structural nonlinearity. It does **not** rule out arbitrary nonlinear C^2 conjugacies of the kind studied in local dynamical-systems linearization theory.*

Hypothesis (a) records that the projection has genuine dynamical content. The affine-coordinate obstruction is carried by hypothesis (b); if (b) holds without (a), the same coordinate conclusion holds, but it is no longer a statement about lost correlations.

3.2 Preliminary Lemma

Lemma 3.2 (Parallel fields on flat connections)**.** *Let (M, ∇) be a connected Banach manifold equipped with a flat affine connection ($R^\nabla = 0$). If a $(1,1)$-tensor field S on M satisfies $\nabla S = 0$, then S is constant in any affine coordinate chart.*

Proof. Since $R^\nabla = 0$, the exponential map provides local affine coordinates where the Christoffel symbols vanish. In these coordinates, $\nabla S = 0$ reduces to $\partial_i S^j_k = 0$, making components constant. Affine transitions preserve this constancy globally. □

3.3 Proof of Theorem 3.1

Proof. **Step 1: Linear flows are affinely flat.** A linear semigroup $\{\tilde{T}_t\}$ with generator $\tilde{A}$ satisfies $\tilde{A}(y) = \tilde{A}y$. The canonical flat connection $\tilde{\nabla}$ on Y has $R^{\tilde{\nabla}} = 0$, and $\tilde{\nabla}^2\tilde{A} = 0$.

Step 2: Affine-coordinate linearity transfers flatness. Suppose an admissible affine chart preserving the induced affine connection represents S_t by a linear semigroup $\tilde{T}_t$. The pullback connection ∇ is flat on M, and $\nabla^2 B = 0$ for the represented generator B.

Step 3: The projected generator has non-constant derivative. The projected generator is $B(m) = D\Pi(m)Am$. Differentiating yields:

$$\nabla_v B = D^2\Pi(m)(Am, v) + D\Pi(m)Av \tag{1}$$

If $\nabla^2 B = 0$, then $\nabla_v B$ would be a parallel tensor field on (M, ∇), and by Lemma 3.2, it would be constant in any affine coordinate chart. But Hypothesis (b) states that $DB(m)$ is non-constant in every such chart. This is a contradiction. □

Scope caveat. Nonconstant Jacobian is not, by itself, an obstruction to all smooth conjugacies to linear normal forms; nonlinear coordinate changes can absorb some nonlinearities under additional spectral hypotheses. The theorem above is an affine-coordinate obstruction statement for the projected generator, not a global no-linearization theorem.

4 Theorem II: Conditional Additive ℓ^1 Diagnostic

Theorem 3.1 demonstrates affine-coordinate nonlinearity produced by the retraction. The next question is how to measure the resulting defect. Theorem 4.2 establishes that, given RE1–RE5, the ℓ^1 norm is the selected diagnostic. The conclusion is conditional on these axioms; relaxing them admits other norms.

4.1 Primitive Axioms for Replica-Extensive Diagnostics

Let E be a finite edge set. For each $e \in E$, let V_e be a finite-dimensional vector space equipped with a faithful local gauge

$$j_e : V_e \to \mathbb{R}_{\geq 0}. \tag{2}$$

A diagnostic is a family

$$J_E : \prod_{e \in E} V_e \to \mathbb{R}_{\geq 0}. \tag{3}$$

We do **not** assume an additive form.

Axiom 1 (RE1 — Global gauge)**.** Each J_E is a faithful convex gauge: $J_E(x) = 0 \iff x = 0$, $J_E(\lambda x) = \lambda J_E(x)$ for $\lambda \geq 0$, and $J_E(x + y) \leq J_E(x) + J_E(y)$.

Axiom 2 (RE2 — Local magnitude sufficiency)**.** There exists a function $A_E : \mathbb{R}^E_{\geq 0} \to \mathbb{R}_{\geq 0}$ such that $J_E(x) = A_E\big((j_e(x_e))_{e \in E}\big)$. Thus the observer sees only independent local magnitudes, not directions or cross-edge phases.

Axiom 3 (RE3 — Disjoint replica additivity)**.** For disjoint finite edge sets E, F: $J_{E \sqcup F}(x \sqcup y) = J_E(x) + J_F(y)$. Equivalently, $A_{E \sqcup F}(u \sqcup v) = A_E(u) + A_F(v)$.

Axiom 4 (RE4 — Edge-exchange symmetry)**.** If $\sigma : E \to F$ is a bijection preserving local gauge type, then $J_F(\sigma x) = J_E(x)$.

Axiom 5 (RE5 — One-edge normalization)**.** For every singleton $E = \{e\}$: $J_{\{e\}}(x_e) = j_e(x_e)$.

4.2 Replica Extensivity Representation Theorem

Theorem 4.1 (Replica Extensivity Representation)**.** *Under RE1–RE5,*

$$J_E(x) = \sum_{e \in E} j_e(x_e). \tag{4}$$

If RE4 is omitted, then $J_E(x) = \sum_{e \in E} c_e j_e(x_e)$ with $c_e > 0$. If RE5 is omitted, the common constant remains undetermined.

Proof. Let $u_e = j_e(x_e)$. By RE2, $J_E(x) = A_E(u)$. Because E is finite, $E = \bigsqcup_{e \in E} \{e\}$. Repeated application of RE3 gives

$$A_E((u_e)_{e \in E}) = \sum_{e \in E} A_{\{e\}}(u_e). \tag{5}$$

For a singleton edge, RE1 gives positive homogeneity: $A_{\{e\}}(\lambda t) = \lambda A_{\{e\}}(t)$ for $\lambda, t \geq 0$. Taking $t = 1$, $A_{\{e\}}(\lambda) = \lambda A_{\{e\}}(1)$. Define $c_e := A_{\{e\}}(1)$. Faithfulness implies $c_e > 0$. Therefore

$$A_E(u) = \sum_{e \in E} c_e u_e. \tag{6}$$

Substituting $u_e = j_e(x_e)$, we get $J_E(x) = \sum_{e \in E} c_e j_e(x_e)$. By RE4, c_e is constant on each edge-exchange orbit. If all edges are equivalent, $c_e = c$. By RE5, $J_{\{e\}}(x_e) = j_e(x_e)$ so $c = 1$. Hence $J_E(x) = \sum_{e \in E} j_e(x_e)$. □

4.3 Coboundary Specialization

Let F be a Banach presheaf over a finite poset P. For each covering edge $e = (a, b)$, define $\delta_e(s) = r_{b,a}s(b) - s(a) \in F(a)$. Set $V_e = F(a)$ and $j_e(v) = \|v\|_{F(a)}$. If the diagnostic family satisfies RE1–RE5, then

$$J_P(s) = J_{E_{\text{cov}}}((\delta_e(s))_e) = \sum_{e \in E_{\text{cov}}} \|\delta_e(s)\|_{F(a)}. \tag{7}$$

Thus the ℓ^1 coboundary norm is derived inside the stated observer class, not from graph structure alone.

4.4 What the theorem does not prove

It does **not** say every reasonable observer is ℓ^1. It says only:

local magnitude sufficiency + disjoint replica additivity + gauge structure + edge symmetry + normalization $\Rightarrow \ell^1$.

4.5 Failure Modes

- **Without RE2:** Cross-edge coupled diagnostics survive: $J(x_1, x_2) = \|x_1 + x_2\|$.
- **Without RE3:** Mixed norms survive: $J(x) = (\sum_e j_e(x_e)^p)^{1/p}$ for $p > 1$.
- **Without RE4:** Weighted ℓ^1 survives: $J(x) = \sum_e c_e j_e(x_e)$.
- **Without RE5:** Only proportional ℓ^1 follows: $J(x) = c \sum_e j_e(x_e)$.
- **Without RE1 homogeneity:** Orlicz sums survive: $J(x) = \sum_e \phi(j_e(x_e))$.

4.6 Position within General Obstruction Framework

The ℓ^1 coboundary norm derived here is not universal across all observers. It is selected within the class of replica-extensive observers satisfying locality, symmetry, and disjoint composability. More generally, an observer is specified by: (aggregation law, admissible transformation monoid, local gauge family). Different choices yield distinct obstruction geometries. For example:

- vector-valued defects with rotational symmetry lead to $\ell^1(E; \ell^2)$,
- quantum systems lead to $\ell^1(E; \text{trace norm})$ only after adding the operational assumptions that select trace distance.

4.6.1 Connection to Observer Passivity and Contextual Congruence

In the general finite obstruction framework [9], observers are formalized as repair-compatible quotient maps $\mathcal{O} : Q^1 \to Q^1_{\mathcal{O}}$, equipped with the induced quotient norm $\Phi_{\mathcal{O}}(y) = \inf_{\mathcal{O}(r)=y} \Phi(r)$. A key structural result (Observer Passivity) is that observers are nonexpansive: $\Phi_{\mathcal{O}}(\mathcal{O}(q)) \leq \Phi(q)$. Observers can erase or identify obstruction, but they cannot create it.

The lift trigger in the general framework is not obstruction mass alone but **context-visible non-congruence**. The general framework provides the categorical machinery to process defects into certified obstruction diagnostics.

The narrowest defensible claim connecting this paper to the general framework is: *the ℓ^1 coboundary norm derived under RE1–RE5 supplies a valid diagnostic for the quotient obstruction Φ in the general finite obstruction calculus, conditional on the observer class specified by the RE axioms.*

5 Theorem III: Finite-Time Density Persistence

5.1 Setup

Consider N sites on a finite graph Λ with local Hilbert spaces ($\dim \mathcal{H}_x \leq d < \infty$), bounded Hamiltonian $H = \sum_e H_e$ with $\|H_e\|_{\text{op}} \leq J$, and retraction $\Pi(\rho) = \bigotimes_x \text{Tr}_{\bar{x}}(\rho)$. Define the commutator:

$$[H_e, \rho_a \otimes \rho_b] = H_e(\rho_a \otimes \rho_b) - (\rho_a \otimes \rho_b)H_e \tag{8}$$

5.2 Explicit Rate Bound

Lemma 5.1 (Local obstruction rate)**.** *For a single bond $e = (a, b)$ with $[H_e, \rho_a \otimes \rho_b] \neq 0$, let $\gamma_e = \|[H_e, \rho_a \otimes \rho_b]\|_1/d^2$. Then $\gamma_e > 0$. For t small:*

$$C_e(t) \geq \gamma_e t - O(J^2 t^2). \tag{9}$$

Proof. On finite-dimensional spaces, the map $t \mapsto \rho(t) - \Pi(\rho(t))$ is analytic. Expanding to first order, the leading term is $-it[H_e, \rho_a \otimes \rho_b]$, whose normalized trace-norm coefficient is $\gamma_e d^2/d^2 = \gamma_e$. The remainder is bounded by $O(J^2 t^2)$. □

5.3 The Density Bound

Theorem 5.2 (Finite-Time Density Persistence)**.** *Let Λ be a bounded-degree finite graph with N sites, finite-dimensional local Hilbert spaces ($\dim \leq d$), and a declared graph-family density bound $|E(\Lambda)| \geq aN$ for some $a > 0$. Let $H = \sum_e H_e$ be a bounded finite-range Hamiltonian with $\|H_e\|_{op} \leq J$, and suppose a fraction $\eta > 0$ of edges satisfy $[H_e, \rho_a \otimes \rho_b] \neq 0$ for an initial product state $\rho(0) = \bigotimes_x \rho_x$. Assume a uniform non-degeneracy condition: $\gamma_e \geq \gamma_{\min} > 0$ for all active edges.*

Then there exist constants $c > 0$ and $t_0 > 0$, depending only on local dimension d, interaction strength J, active fraction η, edge-density bound a, $\gamma_{\min}$, and the bounded-degree/Lieb–Robinson constants (but not on system size N), such that

$$\frac{C(t)}{N} \geq ct \quad \textit{for all } 0 < t \leq t_0, \tag{10}$$

Consequently, on every fixed interval $[t_1, t_2] \subset (0, t_0]$, the defect density is bounded below by the positive constant ct_1.

Proof Sketch. By Lemma 5.2, each active bond e generates a trace-norm defect $C_e(t) \geq \gamma_e t - O(J^2 t^2)$ for small t. By the Lieb–Robinson bound, for times $t < R/v_{\text{LR}}$, the reduced density matrix on any bond is approximated by its isolated evolution up to an exponentially small error.

Summing over at least $\eta|E(\Lambda)|$ active bonds and noting that $\gamma_e \geq \gamma_{\min} > 0$, we obtain $C(t) \geq \eta|E(\Lambda)|(\gamma_{\min} t - KJ^2 t^2)$ for $t \in (0, \gamma_{\min}/(2KJ^2))$. Using $|E(\Lambda)| \geq aN$, dividing by N gives $C(t)/N \geq ct > 0$ on a non-trivial interval, with all constants independent of N. □

6 Worked Examples

6.1 Example 1: Two Qubits (Entangling)

Consider two qubits with $H = J(X \otimes X)$ and initial state $|00\rangle$. The full evolution yields:

$$\rho(t) = \cos^2(Jt)|00\rangle\langle 00| + \sin^2(Jt)|11\rangle\langle 11| + (\text{off-diagonal terms}) \tag{11}$$

The product retraction isolates marginals, yielding:

$$(\Pi\rho)(t) = \cos^4(Jt)|00\rangle\langle 00| + \sin^4(Jt)|11\rangle\langle 11| + \cos^2(Jt)\sin^2(Jt)(|01\rangle\langle 01| + |10\rangle\langle 10|) \tag{12}$$

The trace norm defect $C(t) = \|\rho - \Pi\rho\|_1 > 0$ for $t > 0$, quantifying the failure of the true state to be written as a tensor product of marginals.

6.2 Example 2: Transverse-Field Ising Model

The 1D Ising model $H = -J \sum_i Z_i Z_{i+1} - h \sum_i X_i$ with $|+\rangle^{\otimes N}$ produces strictly positive initial rates $C_e(t) = \gamma_e t + O(J^2 t^2) > 0$. By translational invariance, per-bond density is insensitive to N for $t \ll N/v_{\mathrm{LR}}$.

7 Discussion

This work isolates a structural feature common to projection-based approximations: restricting a linear many-body evolution to a nonlinear manifold introduces a detectable coordinate-level defect. Importantly, these statements are finite-dimensional and finite-time, and rely on explicit structural assumptions.

7.1 Relation to the General Finite Obstruction Framework

The coboundary defects analyzed here embed into the general finite obstruction calculus [9] as follows. The presheaf coboundary $\delta_e(s) = r_{b,a}(s(b)) - s(a)$ provides the raw C^1 data. The quotient obstruction space $Q^1 = C^1/\operatorname{im}(d_0)$ and quotient norm $\Phi([\delta]) = \inf_\alpha \|\delta - d_0\alpha\|_1$ are exactly the objects studied in the general framework. The general framework adds three layers absent from this paper: (1) Observer passivity theory, (2) Contextual congruence and licensed lifts, (3) Reflective completion.

A Appendix B: Bound Shape for Theorem 5.1

This appendix records the constant form used by Theorem 5.1. Let K_d be the finite-dimensional Taylor remainder constant, and let $C_{\mathrm{LR}}, \mu, v_{\mathrm{LR}}$ be Lieb–Robinson constants. The finite-size lower-bound shape is

$$\frac{C(t)}{N} \geq a\eta \left(\gamma_{\min} t - K_d J^2 t^2 - C_{\mathrm{LR}} \exp[-\mu(R_* - v_{\mathrm{LR}} t)]\right). \tag{13}$$

On any finite local window where $K_d J^2 t \leq \frac{\gamma_{\min}}{4}$ and $\varepsilon_{\mathrm{LR}}(t, R_*) \leq \frac{\gamma_{\min} t}{4}$, one obtains the explicit linear lower bound

$$\frac{C(t)}{N} \geq \frac{a\eta\gamma_{\min}}{2}\, t. \tag{14}$$

Acknowledgments

No external funding. No conflicts of interest.

References

1. E. H. Lieb and D. W. Robinson, "The Finite Group Velocity of Quantum Spin Systems," *Commun. Math. Phys.* **28** (1972), 251–257.

2. S. Mac Lane, *Categories for the Working Mathematician*, Springer, 1971.

3. J. Klassen and B. M. Terhal, "Two-local qubit Hamiltonians: when are they stoquastic?," *Quantum* **3** (2019), 139.

4. B. Nachtergaele and R. Sims, "Lieb–Robinson Bounds and the Exponential Clustering Theorem," *Commun. Math. Phys.* **265** (2006), 119–130.

5. S. Bravyi, M. B. Hastings, and F. Verstraete, "Lieb–Robinson Bounds and the Generation of Correlations and Topological Quantum Order," *Phys. Rev. Lett.* **97** (2006), 050401.

6. J. H. Carroll, "Primitive Axiomatics of Finite Cohomological Obstruction Theory," Zenodo, 2026. https://zenodo.org/records/20172395

Changelog

v5.0 (May 2026): Retitled the paper around conditional ℓ^1 diagnostics. Added status, scope, non-claim, and dependency metadata. Added Appendix B.
v4.0 (May 2026): Added §4.6 and §7.1.
v3.0 (April 2026): Audit-hardened revision with explicit scope caveats on Theorem 3.1. RE axioms formalized as RE1–RE5.